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Reproducing and Extending Token Condensation as Adaptation

Training-Free Test-Time Adaptation with Token Condensation

Seminar Paper

in the context of the seminar “Project Representation Learning”

at Friedrich-Alexander-Universität Erlangen-Nürnberg
at the Department Artificial Intelligence in Biomedical Engineering (AIBE)
Image Data Exploration and Analysis (IDEA) Lab

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Submission:	Thursday 8th October 2026

Abstract

This project reproduces and extends Token Condensation as Adaptation (TCA), a training-free test-time adaptation method for CLIP-based Vision Transformers. TCA reduces redundant visual tokens during inference while preserving informative visual evidence, with the goal of improving robustness under distribution shift while reducing computational cost.

The reproduction evaluated CLIP, TDA, EViT, ToMe, and TCA using cross-dataset classification and CIFAR-100-C corruption benchmarks. The reproduced cross-dataset results were close to the values reported in the original paper, with TCA achieving the highest reproduced average accuracy of 68.30%. On selected CIFAR-100-C corruptions, TCA also slightly improved the overall mean accuracy compared with the CLIP baseline.

The project was further extended by testing three token-reduction configurations: Ours-0.035, Ours-0.105, and Ours-0.175, on contrast, snow, and brightness corruptions across five severity levels. Increasing token reduction reduced computational cost from 15.710 to 8.826 GFLOPs, but mean corrupted-data accuracy decreased from 58.64% to 34.71%. The medium configuration, Ours-0.105, achieved a 33.55% computational saving while retaining substantially more accuracy than the most aggressive setting.

Overall, the results show that stronger token condensation improves computational efficiency, but excessive token reduction significantly reduces robustness, especially under stronger distribution shifts. Therefore, token reduction should be selected according to the required balance between efficiency and robustness rather than simply maximizing computational savings.

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1 Introduction

Vision-language models like CLIP show excellent results in many tasks related to visual recognition, however, under changing test distribution, their accuracy decreases. Test-time adaptation (TTA) can improve the performance of vision-language models under changing test distribution.

While building robustness into Vision Transformers (ViTs), it is important to consider their computational efficiency as well. The idea behind ViTs is to process images as a sequence of visual tokens; however, some visual tokens are redundant. In order to solve the problem, Token Condensation as Adaptation (TCA) was proposed. This method removes redundant visual tokens during inference time and keeps only informative visual tokens. TCA is a training-free method that does not involve any updates or backpropagation. That makes TCA relevant in situations when robustness and computational efficiency are required.

In this work, two objectives are pursued: first, to reproduce selected TCA experiments, namely cross-dataset classification, CIFAR-100-C robustness evaluation, and computational cost estimation. TCA is compared to the CLIP baseline and token reduction techniques: EViT and ToMe.

Secondly, the original study is expanded by considering the influence of stronger token reduction on the trade-off between robustness and computational efficiency. Three configurations of TCA are investigated: Ours-0.035, Ours-0.105, and Ours-0.175 on the corrupted data with three types of corruption: contrast, snow, and brightness, varying in the severity level. Both the classification accuracy and computational cost in GFLOPs are estimated.

Research Question: How does increasing the strength of token reduction affect the trade-off between computational efficiency and classification robustness under corrupted input conditions?

The question aims to reveal whether stronger token reduction always provides a better balance of the two characteristics or if a more moderate configuration should be used.

2 Related Work

The main work reproduced in this project is Token Condensation as Adaptation (TCA), proposed by Wang et al. (2025). TCA is a training-free test-time adaptation method for vision-language models under distribution shift. Instead of updating model parameters during inference, it reduces redundant visual tokens while preserving useful information for classification. TCA uses domain-aware token information and logit correction to combine adaptation with lower computational cost. The original work reports improved performance on cross-dataset and corrupted-image benchmarks while requiring less computation.

Several previous methods also address test-time adaptation for vision-language models. Karmanov et al. (2024) introduced the Training-free Dynamic Adapter (TDA), which uses a lightweight key-value cache of test features and pseudo labels to refine predictions without backpropagation. Test-Time Prompt Tuning (TPT), proposed by Shu et al. (2022), instead adapts CLIP's textual prompt for each test sample by minimizing prediction entropy across augmented views. Unlike TCA and TDA, TPT requires optimization during inference and therefore introduces additional computational cost.

A related research direction focuses on reducing the number of visual tokens processed by Vision Transformers. EViT (Liang et al., 2022) identifies important tokens using attention information and reorganizes less informative ones. Token Merging (ToMe) (Bolya et al., 2023) reduces computation by progressively merging similar tokens rather than simply removing them. Adaptive Token Sampling (ATS) (Fayyaz et al., 2022) selects tokens according to their importance and allows the number of retained tokens to vary between inputs. A-ViT (Yin et al., 2022) uses adaptive halting so that less informative tokens can stop being processed at earlier transformer layers.

Overall, TDA and TPT mainly focus on adapting vision-language models to changing test distributions, whereas EViT, ToMe, ATS, and A-ViT mainly focus on reducing visual-token computation. TCA connects these two directions by using token condensation as part of a training-free adaptation strategy. However, stronger token reduction may improve efficiency while removing information needed for robust classification. This trade-off motivates the additional experiment in this project, which compares different TCA token-reduction strengths under corrupted input conditions.

3 Method Summary

TCA stands for Token Condensation as Adaptation and represents a test-time adaptation technique used for models that use Vision Transformers, specifically CLIP. The basic premise behind TCA is a reduction of redundant visual tokens during inference, and retention of tokens needed for the classification task. Thus, using TCA, a model can adapt its token representations without having to update its parameters.

When a Vision Transformer model works with an input image, the image is broken down into patches and these patches become visual tokens, which are then passed through a number of transformer layers. However, some tokens do not necessarily contribute to the prediction and processing of too many tokens is computationally costly. TCA solves this issue by gradually reducing the number of visual tokens processed by the model.

One of the key aspects of TCA is domain-aware token reservoir, which is supposed to help preserve the most important information coming from the input distribution, and not condense all visual tokens, regardless of their importance.

Furthermore, TCA condenses tokens across attention heads. Different heads may assign importance to different visual tokens. Instead of looking at each head individually, TCA takes information about the tokens selected by different attention heads and decides which of them needs to be preserved and which need to be condensed. In other words, the number of tokens used further on in the network can be reduced.

An important property of TCA is that the adaptation happens during inference and without any training or parameter updates. Namely, TCA does not modify and does not train the model, whose parameters are unchanged. Thus, computational benefit of TCA mainly comes from reducing the number of visual-token operations and not from changing the model parameters itself.

The intensity of token reduction affects the tradeoff between information preservation and computational efficiency. The lesser reduction results in more visual information being preserved and in smaller computation reduction. Stronger reduction helps to reduce computation even further, but the price is removing information needed for correct classification. The tradeoff between accuracy and efficiency is what will be the subject of the experimental extension in this project.

4 Reproduction

4.1 Experimental Setup

The reproduction experiments were conducted on the TinyGPU cluster provided by NHR@FAU. GPU resources were allocated through the Slurm workload manager, and the experiments were executed using the authors' publicly available TCA implementation. The experiments were inference-only, since TCA is a training-free adaptation method and therefore does not require additional model training.

Following the reproduction specification, CLIP ViT-B/16 was used as the visual backbone for all experiments. The cross-dataset benchmark consisted of ten classification datasets: Oxford Flowers, Caltech101, DTD, EuroSAT, FGVC Aircraft, Food101, Oxford-IIIT Pets, Stanford Cars, SUN397, and UCF101.

The evaluated methods included the CLIP zero-shot baseline, TDA, two token pruning baselines (EViT and ToMe), and TCA. For the token-pruning comparison, EViT and ToMe were evaluated with a pruning rate of 0.1. For TCA, the corresponding pruning configuration Ours-0.035 was used, following the repository recommendation for approximately comparable computational cost.

In addition to the cross-dataset benchmark, corruption robustness was evaluated using CIFAR-100-C. Three corruption types—contrast, snow, and brightness—were tested at severity levels 1 through 5. Each severity level contained 10,000 test images. The experiments were executed without model fine-tuning, and the final top-1 classification accuracy was recorded for each configuration.

4.2 Cross-Dataset Benchmark

The cross-dataset benchmark evaluates the ability of each method to generalize across ten image classification datasets without additional training. The average top-1 classification accuracy was computed over all ten datasets.

Table 1 summarizes the reproduced results. The CLIP zero-shot baseline achieved an average accuracy of 65.65%, while TDA improved the average to 68.13%. The token-pruning baselines EViT and ToMe achieved 65.16% and 64.84%, respectively.

TCA achieved an average accuracy of 68.30%, outperforming the CLIP zero-shot baseline by 2.65 percentage points. Its average accuracy was also slightly higher than TDA, which achieved 68.13%. In contrast, the EViT and ToMe pruning baselines obtained

Table 1 Reproduced cross-dataset average accuracy.

Method	Average Accuracy (%)
CLIP	65.65
TDA	68.13
EViT	65.16
ToMe	64.84
TCA	68.30

65.16% and 64.84%, respectively. These results show that the TCA configuration maintained competitive classification accuracy while operating with reduced computational cost through token pruning.

4.3 Token Pruning Baselines

EViT and ToMe were selected as baseline token pruning algorithms for comparison of the proposed TCA approach with existing ones used in visual-token reduction. Both algorithms were tested using a pruning ratio of 0.1, as suggested by the repository for comparison with the TCA setting Ours-0.035.

EViT obtained an average accuracy of 65.16%, whereas ToMe scored 64.84%. Both numbers were lower than those attained by TCA (68.30%), indicating that the type of token pruning strategy made a significant difference in classification results.

4.4 CIFAR-100-C Robustness

To evaluate robustness under distribution shift, CLIP and TCA were tested on three CIFAR-100-C corruption types: contrast, snow, and brightness. Each corruption was evaluated at severity levels 1 through 5, where higher severity corresponds to stronger corruption.

Table 2 reports the top-1 classification accuracy for both methods. TCA achieved higher accuracy than CLIP in 14 of the 15 tested corruption–severity settings. The largest improvements were observed for contrast corruption, particularly at severity level 3, where TCA improved accuracy from 59.89% to 61.15%.

Across all five severity levels, TCA improved the mean accuracy for contrast from 55.76% to 56.64%, for snow from 54.65% to 55.11%, and for brightness from 63.89% to 64.17%. The overall average across the 15 corruption settings increased from 58.10% for CLIP to 58.64% for TCA. The only tested setting in which TCA was slightly below CLIP was brightness at severity 5, where the accuracy decreased by 0.15 percentage

Table 2 CIFAR-100-C robustness results for CLIP and TCA.

Corruption	Severity	CLIP (%)	TCA (%)
Contrast	1	67.43	68.30
Contrast	2	63.28	64.34
Contrast	3	59.89	61.15
Contrast	4	54.00	54.92
Contrast	5	34.19	34.50
Snow	1	62.37	62.86
Snow	2	56.28	56.53
Snow	3	54.62	54.94
Snow	4	51.52	52.20
Snow	5	48.44	49.04
Brightness	1	67.57	68.13
Brightness	2	66.66	66.95
Brightness	3	65.16	65.32
Brightness	4	62.90	63.42
Brightness	5	57.18	57.03

points.

Figure 1 shows the mean accuracy across the three corruption types at each severity level.

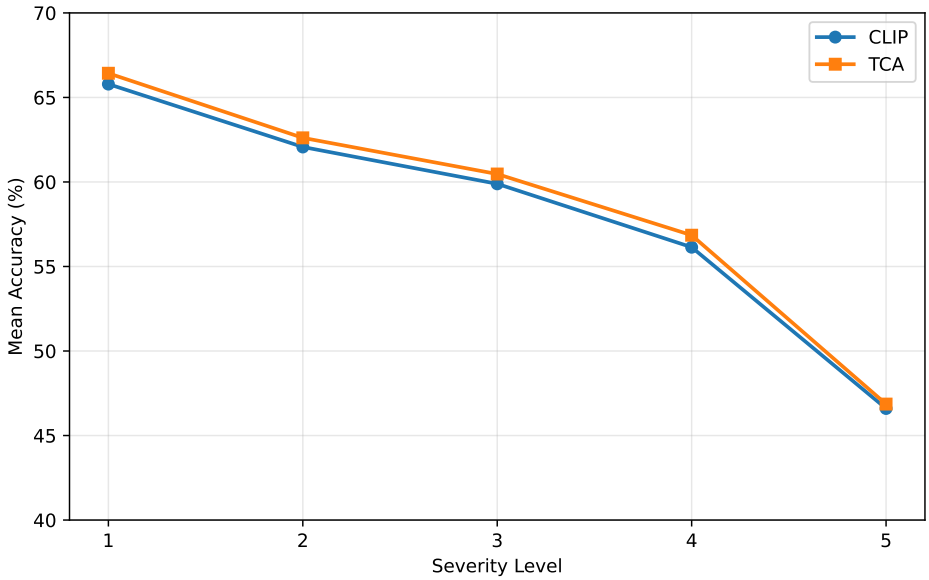


Figure 1 Mean CIFAR-100-C accuracy across contrast, snow, and brightness corruptions at severity levels 1–5.

4.5 Computational Cost

The computational cost of the baseline and token-pruning methods was estimated using FLOP profiling on the CLIP ViT-B/16 visual encoder. Table 3 shows the measured

computational cost for the evaluated configurations.

Table 3 Measured computational cost of the evaluated methods.

Method	GFLOPs
CLIP baseline	17.582
EViT-0.1	15.412
ToMe-0.1	15.304
TCA (Ours-0.035)	15.710

Relative to the CLIP baseline, EViT reduced the measured computation by approximately 12.34%, ToMe by 12.96%, and TCA by 10.65%. Thus, TCA reduced the visual-encoder computation while maintaining a substantially higher cross-dataset accuracy than the two token-pruning baselines in the reproduced experiments.

4.6 Discussion of Deviations

The reproduced cross-dataset results were compared with the reference values reported in the paper. Table 4 summarizes the differences between the reported and reproduced average accuracies.

Table 4 Comparison between reported and reproduced cross-dataset results.

Method	Reported (%)	Reproduced (%)	Difference	Rel. Dev. (%)
CLIP	64.59	65.65	+1.06	+1.64
TDA	67.53	68.13	+0.60	+0.89
TCA	68.69	68.30	-0.39	-0.57

Figure 2 provides a visual comparison between the reported and reproduced cross-dataset average accuracies.

All three reproduced average accuracies were within 2% of the corresponding reported values. CLIP showed the largest relative deviation, at approximately 1.64%, while TDA differed by 0.89% and TCA by only 0.57%. Therefore, the main cross-dataset reproduction satisfies the specified acceptance criterion.

The computational-cost measurements were also close to the values reported in the paper. The reported CLIP cost was 17.59 GFLOPs, compared with 17.582 GFLOPs in the reproduction. For TCA, the reported cost was 15.45 GFLOPs, while the measured value was 15.710 GFLOPs. This corresponds to a relative deviation of approximately 1.68% for TCA.

Small differences between the reproduced and reported results can arise from implementation details, software-library versions, numerical precision, dataset preprocess-

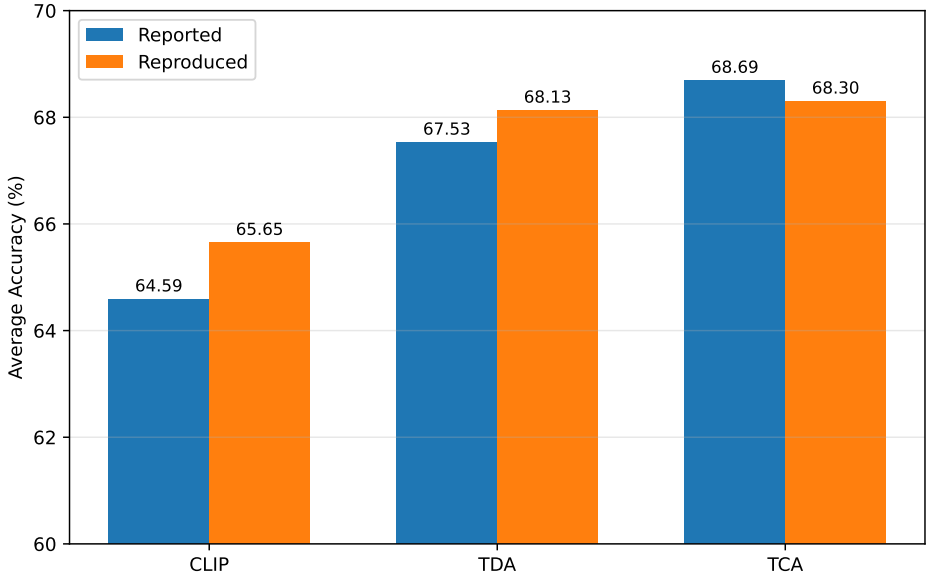


Figure 2 Comparison of reported and reproduced cross-dataset average accuracy.

ing, and the initialization of the TCA reservoir. The assignment specification also notes that TCA may show minor variation because of reservoir initialization.

The FLOP measurements should additionally be interpreted as approximate profiling results. During profiling, the FLOP-counting tool reported several unsupported operators. Consequently, some auxiliary operations may not be fully represented in the measured totals. Nevertheless, the reproduced values remain close to the reference values and show the expected reduction in computation for TCA relative to the unpruned CLIP baseline.

5 Own Experiment

5.1 Motivation

Results from the replication experiments revealed that TCA could minimize the cost of computation without compromising the accuracy of classifications. In particular, the Ours-0.035 setting minimized the cost of computation relative to the CLIP model but maintained higher accuracy in the cross-dataset and corruption benchmarks evaluated.

Nevertheless, a critical question arises from the above findings: what would happen if the amount of token reduction is increased? The increase in the number of tokens would minimize the cost of computation, but it would also eliminate useful visual features resulting in low accuracy of classification. As a result, another set of experiments was designed to establish the relationship between computational efficiency and classification performance.

The new set of experiments involves comparing the performance of three different settings of TCA: Ours-0.035, Ours-0.105, and Ours-175. These settings depict an increase in the number of tokens that were reduced. The performance of these settings was compared using CIFAR-100-C with contrast, snow, and brightness corruptions at severity levels 1-5.

5.2 Hypothesis

From the results that have been reproduced above, it is clear that an increase in the token reduction intensity should bring down the computational cost while at the same time reducing classification accuracy. This effect will be more pronounced with higher levels of image corruption. Consequently, a lower token reduction intensity would yield a better balance than a higher token reduction intensity.

5.3 Experimental Setup

The own experiment was conducted using the same CLIP ViT-B/16 backbone and TCA implementation used in the reproduction experiments. The experiments were executed on the NHR@FAU TinyGPU cluster using the Slurm workload manager.

Three TCA configurations were considered: Ours-0.035, Ours-0.105, and Ours-0.175. Ours-0.035 corresponds to the less aggressive configuration evaluated during the reproduction, while Ours-0.105 and Ours-0.175 were used to investigate progressively stronger token reduction.

Robustness was evaluated on CIFAR-100-C using three corruption types: contrast, snow, and brightness. For each corruption type, five severity levels (1–5) were evaluated. This gives 15 corruption–severity combinations for each TCA configuration.

For the additional experiment, Ours-0.105 and Ours-0.175 were evaluated on all 15 combinations, resulting in 30 additional evaluation runs. The Ours-0.035 results from the reproduction were used as the reference configuration for comparison.

For each configuration, classification accuracy was recorded for the corruption experiments. Computational cost was evaluated separately in GFLOPs. The measured costs were 15.710 GFLOPs for Ours-0.035, 11.683 GFLOPs for Ours-0.105, and 8.826 GFLOPs for Ours-0.175. These measurements allow the effect of stronger token reduction on both robustness and computational efficiency to be compared.

5.4 Results

Table 5 summarizes the overall accuracy and computational cost for the three TCA configurations. Increasing the token reduction strength consistently reduced the computational cost, but it also resulted in lower classification accuracy.

Table 5 Accuracy and computational cost for the tested TCA configurations.

Configuration	Mean Accuracy (%)	GFLOPs	Compute Saving (%)
Ours-0.035	58.64	15.710	10.65
Ours-0.105	52.33	11.683	33.55
Ours-0.175	34.71	8.826	49.80

Ours-0.035 achieved the highest mean accuracy of 58.64%, with a computational cost of 15.710 GFLOPs. Increasing the token reduction strength to Ours-0.105 reduced the computational cost to 11.683 GFLOPs, corresponding to a compute saving of 33.55%, while the mean accuracy decreased to 52.33%.

The strongest setting, Ours-0.175, achieved the lowest computational cost of 8.826 GFLOPs and the largest compute saving of 49.80%. However, its mean accuracy decreased substantially to 34.71%.

The effect of corruption severity was also visible across the experiments. Accuracy generally decreased as the severity level increased from 1 to 5. This decrease was more pronounced for the stronger token reduction settings, particularly Ours-0.175.

5.5 Discussion

The findings confirm the hypothesis of a trade-off between the reduction strength and efficiency of the model versus classification accuracy. With the increase of the reduction strength, the computational cost of the model significantly decreased. At the same time, it lost its robustness to corrupted images.

In comparison of the two approaches, a moderate trade-off can be seen in the change from Ours-0.035 to Ours-0.105. The computational cost was reduced from 15.710 to 11.683 GFLOPs, whereas the mean accuracy went down from 58.64% to 52.33%. The approach Ours-0.175 managed to decrease the computational cost even more to 8.826 GFLOPs, yet it showed very low mean accuracy of 34.71%.

This contrast becomes especially relevant in case of the corruption of the images with increasing severity. The stronger approaches have less visual tokens retained to process the input image. Therefore, under the corrupted conditions, the aggressive reduction of the tokens may exclude some useful information for classification. This might be the reason of a larger accuracy drop of Ours-0.175.

Out of two stronger configurations in an additional experiment, Ours-0.105 shows a better balance of efficiency and accuracy in comparison with Ours-0.175. It allows achieving a computational saving of 33.55%, while maintaining much higher accuracy. In turn, maximum computational saving of Ours-0.175 leads to a significant performance degradation.

Thus, one can conclude that the token reduction strength should not be chosen only according to computational cost minimization. The optimal configuration depends on the trade-off between efficiency and robustness. In case the performance under distribution shift is important, the moderate token reduction is preferable.

6 Conclusion

This study reproduced and expanded upon selected experiments of Token Condensation as Adaptation (TCA). The reproduction involved testing of TCA as well as the CLIP baseline and the token reduction approach in cross-dataset and CIFAR-100-C settings. The reproduced results matched the reported results well and indicated that TCA allows one to lower the computational costs without compromising the classification accuracy too much.

The second part of the project further extended the evaluation by studying the impact of different token reduction strengths. Three different TCA setups – Ours-0.035, Ours-0.105, and Ours-0.175 – were evaluated using contrast, snow, and brightness corruption sets with 5 severity levels each. Raising the token reduction strength allowed to reduce the computational cost from 15.710 GFLOPs for Ours-0.035 to 11.683 GFLOPs for Ours-0.105 and 8.826 GFLOPs for Ours-0.175. At the same time, the accuracy became significantly lower with the respective mean values equaling 58.64%, 52.33%, and 34.71%.

These results indicate a key trade-off between efficiency and robustness. From the two stronger TCA setups, Ours-0.105 demonstrated a better trade-off than Ours-0.175 since it allows to achieve significant efficiency gain without a significant drop in accuracy as opposed to the case of Ours-0.175.

This experiment only covered three types of CIFAR-100-C corruption with a limited number of token reduction strengths. In future, one may try to use more corruption types, datasets, and token reduction strengths in order to check if the same trade-off would appear.

In general, these results suggest that one cannot try to get the lowest computational cost by simply maximizing the token reduction strength. Instead, the proper level should be chosen depending on the required trade-off between the efficiency and robustness.

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GitHub Link

For reproducibility, the code, experiment scripts, configuration files, and results produced during this project are publicly available in the GitHub repository:

`https://github.com/sayeda077/TCA-Reproduction-and-Extension`